

Attributes

- **Attribute (or dimensions, features, variables)**

- A data field, representing a characteristic or feature of a data object.

- *E.g., customer_ID, name, address*

- **Types:**

- Nominal (e.g., red, blue)

- Binary (e.g., {true, false})

- Ordinal (e.g., {freshman, sophomore, junior, senior})

- Numeric: quantitative

- Interval-scaled: 100°C is interval scales

- Ratio-scaled: 100°K is ratio scaled since it is twice as high as 50°K

- Discrete vs. Continuous Attributes

Attribute Types

- ❑ **Nominal:** categories, states, or “names of things”

- ❑ *Hair_color* = {*auburn, black, blond, brown, grey, red, white*}
 - ❑ marital status, occupation, ID numbers, zip codes

- ❑ **Binary**

- ❑ Nominal attribute with only 2 states (0 and 1)
 - ❑ Symmetric binary: both outcomes equally important
 - ❑ e.g., gender
 - ❑ Asymmetric binary: outcomes not equally important.
 - ❑ e.g., medical test (positive vs. negative)
 - ❑ Convention: assign 1 to most important outcome (e.g., HIV positive)

- ❑ **Ordinal**

- ❑ Values have a meaningful order (ranking) but magnitude between successive values is not known
 - ❑ *Size* = {*small, medium, large*}, grades, army rankings